



CS 630: Speech Technology

Course Code	CS 630
Course Name	Speech Technology
L-T-P-C	3-0-2-4
Intended for	B.Tech
Distribution	Elective
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Overview of Speech Technology:** What is Speech Technology? Why is it important? Its applications and issues.
- **Speech Production:** Mechanism of speech production; Categories of sounds; Sound units in indian languages.
- **Nature of Speech Signal:** Source-system characteristics; Segmental and suprasegmental features; Temporal and spectral parameters for sound units in indian languages.
- **Basics of Digital Signal Processing:** Signals and systems; Discrete fourier transform; Digital filtering; Stochastic processes.
- **Speech Signal Processing Methods:** Short-time spectrum analysis; Spectrograms; Linear prediction analysis; Cepstrum analysis.
- **Speech Recognition:** Isolated word recognition; Connected word recognition Continuous
- **Speech Recognition:** Speech recognition problem; Hidden markov models.
- **Other Applications:** Word spotting; Speaker recognition; Speech enhancement; Speech synthesis; Practical issues in speech technology.

Text Books

1. L R Rabiner and R W Schafer, **Theory and Application of Digital Speech Processing**, Pearson, 2011.
2. L R Rabiner, B-H Juang and B Yegnanarayana, **Fundamentals of Speech Recognition**, Pearson, 2009.
3. Xuedong Huang, Alex Acero, Hsiao-wuen Hon, **Spoken Language Processing: A guide to Theory, Algorithm, and System Development**, Prentice Hall, 2001.

References

1. Oppenheim and Schafer, **Discrete-Time Signal Processing**, PHI, 2001.
2. T W Parsons, **Voice and Speech Processing**, McGraw Hill, 1986.
3. Thomas Quatieri, **Discrete-time Speech Processing: Principles and Practice**, PH, 2001.
4. Rabiner and Schafer, **Digital Processing of Speech Signals**, Pearson Education, 1993.
5. Douglas O Shaughnessy, **Speech Communications**, University Press, 2001.